

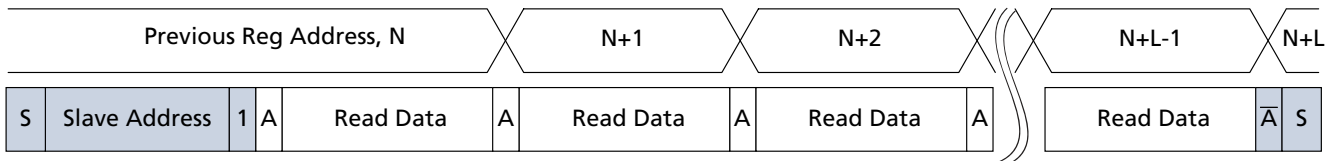


MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Two-Wire Serial Interface

Sequential READ, Start from Current Location

This sequence (Figure 26) starts in the same way as the single READ from current location (Figure 24 on page 40). Instead of generating a no-acknowledge bit after the first byte of data has been transferred, the master generates an acknowledge bit, and continues to perform byte reads until “L” bytes have been read.

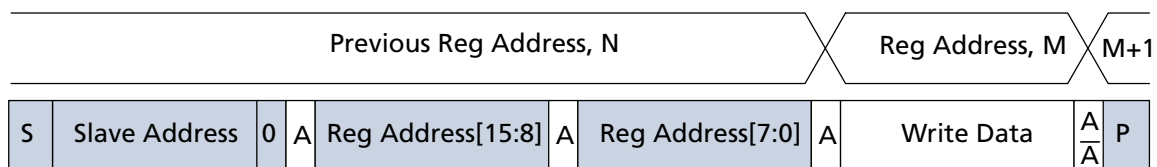
Figure 26: Sequential READ, Start from Current Location



Single WRITE to Random Location

This sequence (Figure 27) begins with the master generating a start condition. The slave address/data direction byte signals a WRITE and is followed by the high then low bytes of the register address that is to be written. The master follows this with the byte of WRITE data. The WRITE is terminated by the master generating a stop condition.

Figure 27: Single WRITE to Random Location



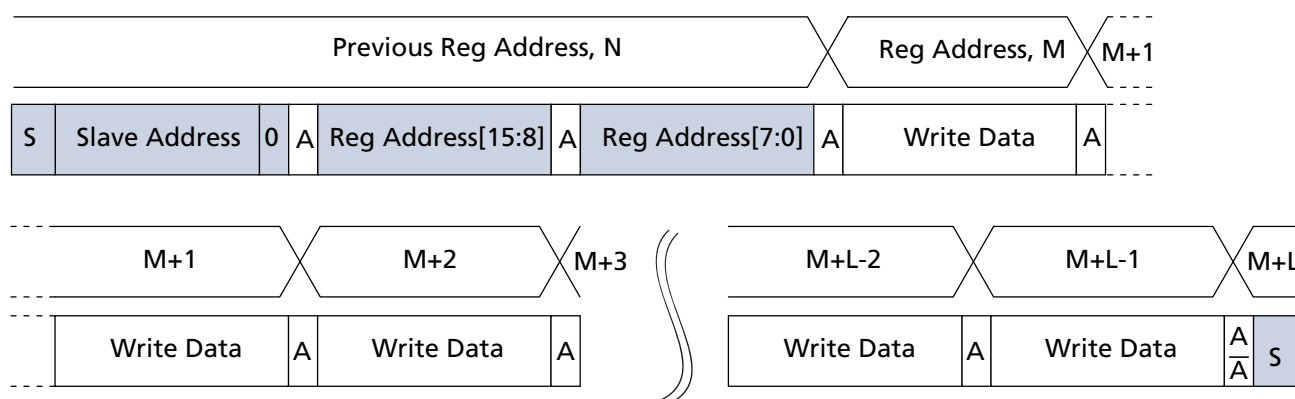


MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Two-Wire Serial Interface

Sequential WRITE, Start at Random Location

This sequence (Figure 28) starts in the same way as the single WRITE to random location (Figure 27 on page 41). Instead of generating a no-acknowledge bit after the first byte of data has been transferred, the master generates an acknowledge bit, and continues to perform byte writes until “L” bytes have been written. The WRITE is terminated by the master generating a stop condition.

Figure 28: Sequential WRITE, Start at Random Location





MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Registers and Variables

Registers and Variables

Four types of configuration controls are available:

- Hardware registers
- Driver variables
- Special function registers
- MCU SRAM

The following convention is used in the text below to designate registers and variables:

R0x3012 or R0x3012[3:0]

These refer to two-wire accessible register number 3012. [3:0] indicate bits. Registers numbers range from 0x0000 through 0xFFFF and bits range from 15 through 0. Not all register numbers are used.

- ae.Target
This refers to variable "Target" in the AE driver.
- SFR 0x1080 or SRAM 0x0400
This refers to special function register or SRAM located at address 0x1080 in MCU memory space.

How to Access Registers and Variables

Registers, variables, and SFRs are accessed in different ways.

Registers

Hardware registers grouped internally by pages.

R0x3000 – R0x31FE = Sensor Core Page 0

R0x3200 – R0x33FE = SOC Page 1

R0x3400 – R0x35BE = SOC Page 2

- Page 0 contains sensor controls.
- Page 1 contains color pipeline controls.
- Page 2 contains MIPI, output FIFO and more color pipeline controls.

Not all the registers in each page are used, not all the bits in each register might be used. Registers are accessed by serial interface READ and WRITE consisting of a 16-bit address and 16-bit data.

Variables

Variables are located in the microcontroller RAM memory. Each driver, such as auto exposure, white balance, and auto focus, has a unique driver ID (from 0 through 31) and a set of public variables organized as a structure. Each variable in this structure is uniquely identified by its offset from the top of the structure and its size. The size can be 1 or 2 bytes, while the offset is 1 byte.

All driver variables (public and private) can be accessed via R0x098C and R0x0990. While two access modes are available-access by physical address and by logical address, the public variables are typically accessed by the logical method. The logical address, which is set in R0x098C, consists of a 5-bit driver ID number and a variable offset. Examples are provided below.



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To set the variable ae.Target = 0x0050:

- The variable has a driver ID of 2. Therefore, set R0x098C[12:8] = 0x0002.
- The variable has an offset of 6. Therefore, set R0x098C[7:0] = 0x0006.
- This is a logical access. Therefore, set R0x098C[14:13] = 01.
- The size of the variable is 8 bits. Therefore, set R0x098C[15] = 0x0001.
- By combining these bits, R0x098C = 0xA206.
- Set R0x0990 = 0x0050 for the value of the variable.

To read the variable ae.Target:

- Since this is the same variable as the above example, R0x098C = 0xA206.
- Read R0x0990 for the current variable value.

To set the variable mon.arg1 = 0x1234:

- The variable has a driver ID of 0. Therefore, set R0x098C[12:8] = 0.
- The variable has an offset of 3. Therefore, set R0x0990[7:0] = 0x0003.
- This is a logical access. Therefore, set R0x098C[14:13] = 01.
- The size of the variable is 16 bits. Therefore, set R0x098C[15] = 0.
- By combining these bits, R0x098C = 0x2003.
- Set R0x0990 = 0x1234 for the value of the variable.

To read the variable mon.arg1:

- Since this is the same variable as the above example, R0x098C = 0x2003.
- Read R0x0990 for the current variable value.

Special Function Registers and MCU SRAM

Special function registers are registers connected to the local bus of the microcontroller. These registers include GPIO, waveform generator, and those important for firmware operation. SFR are accessed by physical address. MCU SRAM consists of 1K system memory and 1K user memory. Examples of access:

- Write into user SRAM. Use to upload code
 - a. R0x098C = 0x0400 // address
 - b. R0x0990 = 0x1234 // write 16-bit value
- Read from user SRAM
 - c. R0x098C = 0x0400 // address
 - d. Read R0x0990 // read 16-bit value
- Write to 8-bit GPIO register
 - e. R0x098C = 0x9079 // GPIO_DIR_L at 0x1079
 - f. R0x0990 = 0x00FE // Configure GPIO[0] as output
- Read from 8-bit GPIO register
 - g. R0x098C = 0x9079 // GPIO_DIR_L at 0x1079
 - h. Read R0x0990 // Check GPIO[7:0] pad state



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Core Registers

Double-buffered Registers

Some sensor settings cannot be changed during frame readout. For example, changing `x_addr_end` `R0x3008` part way through frame readout results in inconsistent `LINE_VALID` behavior. To avoid this, the sensor core double buffers many registers by implementing a pending and a live version. READs and WRITEs access the pending register. The live register controls the sensor operation.

The value in the pending register is transferred to a live register at a fixed point in the frame timing, called frame start. Frame start is defined as the point at which the first dark row is read out. By default, this occurs 10 row times before `FRAME_VALID` goes HIGH.

`R0x0248[15]` can be used to inhibit transfers from the pending to the live registers. This control should be used when making many register changes that must take effect simultaneously.

Bad Frames

A bad frame is a frame where all rows do not have the same integration time, or where offsets to the pixel values changed during the frame.

Many changes to the sensor register settings can cause a bad frame. For example, when `x_addr_end` `R0x3008` is changed, the new register value does not affect sensor behavior until the next frame start. However, the frame that would be read out at that frame-start has been integrated using the old row width. Consequently, reading it out using the new row width results in a frame with an incorrect integration time.

By default, most bad frames are masked: `LINE_VALID` and `FRAME_VALID` are inhibited for these frames so that the vertical blanking time between frames is extended by the frame time. This feature can be disabled with `R0x0248[10]`.

Changes to Integration Time

If the integration time is changed while `FRAME_VALID` is asserted for frame n ; the first frame output using the new integration time is frame $(n + 2)$. The sequence is as follows:

1. During frame n , the new integration time is held in the pending register.
2. At the start of frame $(n + 1)$, the new integration time is transferred to the live register. Integration for each row of frame $(n + 1)$ has been completed using the old integration time. The earliest time that a row can start integrating using the new integration time is immediately after that row has been read for frame $(n + 1)$. The actual time that rows start integrating using the new integration time is dependent on the new value of the integration time.
3. When frame $(n + 1)$ is read out, it is integrated using the new integration time. If the integration time is changed on successive frames, each value written is applied to a single frame; the latency between writing a value and it affecting the frame readout remains at two frames.

Changes to Gain Settings

When the gain settings are changed, the gain is usually updated on the next frame start. When the integration time and the gain are changed simultaneously, the gain update is held off by one frame so that the first frame output with the new integration time also has the new gain applied.



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If the gain and integration time are both changed on successive frames, some gain value is overwritten without being applied, while each integration time is used for one single frame.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Summary

Register Summary

Table 10: Sensor Core Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3012	coarse_integration_time	dddd dddd dddd dddd	16 (0x0010)
R0x3014	fine_integration_time	dddd dddd dddd dddd	249 (0x00F9)
R0x301E	data_pedestal	0000 00dd dddd dddd	42 (0x002A)
R0x3028	analog_gain_code_global	0000 0000 0ddd dddd	13 (0x000D)
R0x302A	analog_gain_code_greenr	0000 0000 0ddd dddd	13 (0x000D)
R0x302C	analog_gain_code_red	0000 0000 0ddd dddd	13 (0x000D)
R0x302E	analog_gain_code_blue	0000 0000 0ddd dddd	13 (0x000D)
R0x3030	analog_gain_code_greenb	0000 0000 0ddd dddd	13 (0x000D)
R0x3032	digital_gain_greenr	0000 0ddd 0000 0000	256 (0x0100)
R0x3034	digital_gain_red	0000 0ddd 0000 0000	256 (0x0100)
R0x3036	digital_gain_blue	0000 0ddd 0000 0000	256 (0x0100)
R0x3038	digital_gain_greenb	0000 0ddd 0000 0000	256 (0x0100)
R0x304A	otpm_control	0000 0000 0??? 0???d	0 (0x0000)
R0x304C	otpm_program_value	dddd dddd dddd dddd	0 (0x0000)
R0x304E	otpm_read_value	???? ???? ???? ????d	0 (0x0000)

Table 11: SOC1 Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3210	color_pipeline_control	0000 0d0d d0dd d000	432 (0x01B0)
R0x321C	output_control_status	???? dddd d000 dddd	3 (0x0003)
R0x321E	output_control_status_2	dddd dddd dddd dddd	16 (0x0010)
R0x326C	2d_aperture_parameters	0ddd dddd dddd dddd	4616 (0x1208)
R0x327A	red_offset	0000 000d dddd dddd	42 (0x002A)
R0x327C	green1_offset	0000 000d dddd dddd	42 (0x002A)
R0x327E	green2_offset	0000 000d dddd dddd	42 (0x002A)
R0x3280	blue_offset	0000 000d dddd dddd	42 (0x002A)
R0x328E	dm_edge_th	0000 0000 dddd dddd	12 (0x000C)
R0x3290	test_pattern	0000 0000 0dd0 0000	0 (0x0000)
R0x3330	output_format_test	0000 dddd dddd dddd	0 (0x0000)
R0x337C	yuv_ycbcr_control	0000 0000 0000 dddd	6 (0x0006)
R0x33F4	kernel_config	0000 000d dddd dddd	3 (0x0003)



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Table 12: SOC2 Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3400	mipi_control	dddd dddd dddd ?ddd	30766 (0x782E)
R0x3402	mipi_status	000d d??0 00?? 000?	17 (0x0011)
R0x3404	custom_short_pkt	00?d dddd dd00 0000	0 (0x0000)
R0x3408	line_byte_cnt	dddd dddd dddd dddd	512 (0x0200)
R0x340C	custom_short_pkt_wc	dddd dddd dddd dddd	0 (0x0000)
R0x3580	ae_zone_x	dddd dddd dddd dddd	4864 (0x1300)
R0x3582	ae_zone_y	dddd dddd dddd dddd	3584 (0x0E00)
R0x3584	ae_window_size_lo	dddd dddd dddd dddd	19200 (0x4B00)
R0x3586	ae_window_size_hi	0000 0000 0000 000d	0 (0x0000)
R0x3642	pga_1	0000 0ddd dddd dddd	516 (0x0204)
R0x3644	pga_2	0000 0ddd dddd dddd	644 (0x0284)
R0x364E	pga_3	dddd dddd dddd dddd	16 (0x0010)
R0x3650	pga_4	dddd dddd dddd dddd	0 (0x0000)
R0x3652	pga_5	dddd dddd dddd dddd	0 (0x0000)
R0x3654	pga_6	dddd dddd dddd dddd	0 (0x0000)
R0x3656	pga_7	dddd dddd dddd dddd	0 (0x0000)
R0x3658	pga_8	dddd dddd dddd dddd	16 (0x0010)
R0x365A	pga_9	dddd dddd dddd dddd	0 (0x0000)
R0x365C	pga_10	dddd dddd dddd dddd	0 (0x0000)
R0x365E	pga_11	dddd dddd dddd dddd	0 (0x0000)
R0x3660	pga_12	dddd dddd dddd dddd	0 (0x0000)
R0x3662	pga_13	dddd dddd dddd dddd	16 (0x0010)
R0x3664	pga_14	dddd dddd dddd dddd	0 (0x0000)
R0x3666	pga_15	dddd dddd dddd dddd	0 (0x0000)
R0x3668	pga_16	dddd dddd dddd dddd	0 (0x0000)
R0x366A	pga_17	dddd dddd dddd dddd	0 (0x0000)
R0x366C	pga_18	dddd dddd dddd dddd	16 (0x0010)
R0x366E	pga_19	dddd dddd dddd dddd	0 (0x0000)
R0x3670	pga_20	dddd dddd dddd dddd	0 (0x0000)
R0x3672	pga_21	dddd dddd dddd dddd	0 (0x0000)
R0x3674	pga_22	dddd dddd dddd dddd	0 (0x0000)
R0x3676	pga_23	dddd dddd dddd dddd	0 (0x0000)
R0x3678	pga_24	dddd dddd dddd dddd	0 (0x0000)
R0x367A	pga_25	dddd dddd dddd dddd	0 (0x0000)
R0x367C	pga_26	dddd dddd dddd dddd	0 (0x0000)
R0x367E	pga_27	dddd dddd dddd dddd	0 (0x0000)
R0x3680	pga_28	dddd dddd dddd dddd	0 (0x0000)
R0x3682	pga_29	dddd dddd dddd dddd	0 (0x0000)
R0x3684	pga_30	dddd dddd dddd dddd	0 (0x0000)
R0x3686	pga_31	dddd dddd dddd dddd	0 (0x0000)
R0x3688	pga_32	dddd dddd dddd dddd	0 (0x0000)
R0x368A	pga_33	dddd dddd dddd dddd	0 (0x0000)
R0x368C	pga_34	dddd dddd dddd dddd	0 (0x0000)



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Table 12: SOC2 Registers (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x368E	pga_35	dddd dddd dddd dddd	0 (0x0000)
R0x3690	pga_36	dddd dddd dddd dddd	0 (0x0000)
R0x3692	pga_37	dddd dddd dddd dddd	0 (0x0000)
R0x3694	pga_38	dddd dddd dddd dddd	0 (0x0000)
R0x3696	pga_39	dddd dddd dddd dddd	0 (0x0000)
R0x3698	pga_40	dddd dddd dddd dddd	0 (0x0000)
R0x369A	pga_41	dddd dddd dddd dddd	0 (0x0000)
R0x369C	pga_42	dddd dddd dddd dddd	0 (0x0000)
R0x369E	pga_43	dddd dddd dddd dddd	0 (0x0000)
R0x36A0	pga_44	dddd dddd dddd dddd	0 (0x0000)
R0x36A2	pga_45	dddd dddd dddd dddd	0 (0x0000)
R0x36A4	pga_46	dddd dddd dddd dddd	0 (0x0000)
R0x36A6	pga_47	dddd dddd dddd dddd	0 (0x0000)
R0x36A8	pga_48	dddd dddd dddd dddd	0 (0x0000)
R0x36AA	pga_49	dddd dddd dddd dddd	0 (0x0000)
R0x36AC	pga_50	dddd dddd dddd dddd	0 (0x0000)
R0x36AE	pga_51	dddd dddd dddd dddd	0 (0x0000)
R0x36B0	pga_52	dddd dddd dddd dddd	0 (0x0000)
R0x36B2	pga_53	dddd dddd dddd dddd	0 (0x0000)
R0x36B4	pga_54	dddd dddd dddd dddd	0 (0x0000)
R0x36B6	pga_55	dddd dddd dddd dddd	0 (0x0000)
R0x36B8	pga_56	dddd dddd dddd dddd	0 (0x0000)
R0x36BA	pga_57	dddd dddd dddd dddd	0 (0x0000)
R0x36BC	pga_58	dddd dddd dddd dddd	0 (0x0000)
R0x36BE	pga_59	dddd dddd dddd dddd	0 (0x0000)
R0x36C0	pga_60	dddd dddd dddd dddd	0 (0x0000)
R0x36C2	pga_61	dddd dddd dddd dddd	0 (0x0000)
R0x36C4	pga_62	dddd dddd dddd dddd	0 (0x0000)
R0x36C6	pga_63	dddd dddd dddd dddd	0 (0x0000)
R0x36C8	pga_64	dddd dddd dddd dddd	0 (0x0000)
R0x36CA	pga_65	dddd dddd dddd dddd	0 (0x0000)
R0x36CC	pga_66	dddd dddd dddd dddd	0 (0x0000)
R0x36CE	pga_67	dddd dddd dddd dddd	0 (0x0000)
R0x36D0	pga_68	dddd dddd dddd dddd	0 (0x0000)
R0x36D2	pga_69	dddd dddd dddd dddd	0 (0x0000)
R0x36D4	pga_70	dddd dddd dddd dddd	0 (0x0000)
R0x36D6	pga_71	dddd dddd dddd dddd	0 (0x0000)
R0x36D8	pga_72	dddd dddd dddd dddd	0 (0x0000)
R0x36DA	pga_73	dddd dddd dddd dddd	0 (0x0000)
R0x36DC	pga_74	dddd dddd dddd dddd	0 (0x0000)
R0x36DE	pga_75	dddd dddd dddd dddd	0 (0x0000)
R0x36E0	pga_76	dddd dddd dddd dddd	0 (0x0000)
R0x36E2	pga_77	dddd dddd dddd dddd	0 (0x0000)
R0x36E4	pga_78	dddd dddd dddd dddd	0 (0x0000)



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Table 12: SOC2 Registers (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x36E6	pga_79	dddd dddd dddd dddd	0 (0x0000)
R0x36E8	pga_80	dddd dddd dddd dddd	0 (0x0000)
R0x36EA	pga_81	dddd dddd dddd dddd	0 (0x0000)
R0x36EC	pga_82	dddd dddd dddd dddd	0 (0x0000)
R0x36EE	pga_83	dddd dddd dddd dddd	0 (0x0000)
R0x36F0	pga_84	dddd dddd dddd dddd	0 (0x0000)
R0x36F2	pga_85	dddd dddd dddd dddd	0 (0x0000)
R0x36F4	pga_86	dddd dddd dddd dddd	0 (0x0000)
R0x36F6	pga_87	dddd dddd dddd dddd	0 (0x0000)
R0x36F8	pga_88	dddd dddd dddd dddd	0 (0x0000)
R0x36FA	pga_89	dddd dddd dddd dddd	0 (0x0000)
R0x36FC	pga_90	dddd dddd dddd dddd	0 (0x0000)
R0x36FE	pga_91	dddd dddd dddd dddd	0 (0x0000)
R0x3700	pga_92	dddd dddd dddd dddd	0 (0x0000)
R0x3702	pga_93	dddd dddd dddd dddd	0 (0x0000)
R0x3704	pga_94	dddd dddd dddd dddd	0 (0x0000)
R0x3706	pga_95	dddd dddd dddd dddd	0 (0x0000)
R0x3708	pga_96	dddd dddd dddd dddd	0 (0x0000)
R0x370A	pga_97	dddd dddd dddd dddd	0 (0x0000)
R0x370C	pga_98	dddd dddd dddd dddd	0 (0x0000)
R0x370E	pga_99	dddd dddd dddd dddd	0 (0x0000)
R0x3710	pga_100	dddd dddd dddd dddd	0 (0x0000)
R0x3712	pga_101	dddd dddd dddd dddd	0 (0x0000)
R0x3714	pga_102	dddd dddd dddd dddd	0 (0x0000)



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Table 13: SYSTL Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0000	k24a_chip_id	???? ???? ???? ???? ?	9344 (0x2480)
R0x0010	pll_dividers	00dd dddd dddd dddd	384 (0x0180)
R0x0012	pll_p_dividers	00dd dddd dddd dddd	0 (0x0000)
R0x0014	pll_control	?0dd dddd dddd dddd	8517 (0x2145)
R0x0016	clocks_control	dddd dddd dddd dddd	0 (0x0000)
R0x0018	standby_control_and_status	?d00 0000 00dd dddd	16425 (0x4029)
R0x001A	reset_and_misc_control	0000 00dd 00dd d0dd	16 (0x0010)
R0x001C	mcu_boot_mode	???? ???? dddd dddd	0 (0x0000)
R0x001E	pad_slew	0000 0ddd 0ddd 0ddd	1024 (0x0400)
R0x0022	vdd_dis_counter	dddd dddd dddd dddd	1080 (0x0438)
R0x0024	gpi_status	0000 0000 000? ???? ?	0 (0x0000)
R0x0028	hard_standby_sel	0000 0000 0000 000d	1 (0x0001)

Table 14: XDMA Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x098C	mcu_variable_address	dddd dddd dddd dddd	0 (0x0000)
R0x0990	mcu_variable_data0	dddd dddd dddd dddd	0 (0x0000)
R0x0992	mcu_variable_data1	dddd dddd dddd dddd	0 (0x0000)
R0x0994	mcu_variable_data2	dddd dddd dddd dddd	0 (0x0000)
R0x0996	mcu_variable_data3	dddd dddd dddd dddd	0 (0x0000)
R0x0998	mcu_variable_data4	dddd dddd dddd dddd	0 (0x0000)
R0x099A	mcu_variable_data5	dddd dddd dddd dddd	0 (0x0000)
R0x099C	mcu_variable_data6	dddd dddd dddd dddd	0 (0x0000)
R0x099E	mcu_variable_data7	dddd dddd dddd dddd	0 (0x0000)

Table 15: Driver ID = 1: Sequencer Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0002	seq_mode	0000 0000 dddd dddd	15 (0x000F)
R0x0003	seq_cmd	0000 0000 dddd dddd	0 (0x0000)
R0x0004	seq_state	0000 0000 dddd dddd	3 (0x0003)
R0x0006	seq_flash_type	0000 0000 dddd dddd	0 (0x0000)
R0x0009	seq_ae_fast_buff	0000 0000 dddd dddd	32 (0x0020)
R0x000A	seq_ae_fast_step	0000 0000 dddd dddd	1 (0x0001)
R0x000B	seq_awb_cont_buff	0000 0000 dddd dddd	8 (0x0008)
R0x000C	seq_awb_cont_step	0000 0000 dddd dddd	2 (0x0002)
R0x0011	seq_options	0000 0000 dddd dddd	168 (0x00A8)
R0x0013	seq_flash_th	dddd dddd dddd dddd	0 (0x0000)



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Table 15: Driver ID = 1: Sequencer Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0015	seq_cap_mode	0000 0000 dddd dddd	0 (0x0000)
R0x0016	seq_cap_num_frames	0000 0000 dddd dddd	3 (0x0003)
R0x0017	seq_preview_0_ae	0000 0000 dddd dddd	0 (0x0000)
R0x0018	seq_preview_0_fd	0000 0000 dddd dddd	0 (0x0000)
R0x0019	seq_preview_0_awb	0000 0000 dddd dddd	0 (0x0000)
R0x001A	seq_preview_0_hg	0000 0000 dddd dddd	0 (0x0000)
R0x001B	seq_preview_0_flash	0000 0000 dddd dddd	0 (0x0000)
R0x001C	seq_preview_0_skipframe	0000 0000 dddd dddd	64 (0x0040)
R0x001D	seq_preview_1_ae	0000 0000 dddd dddd	3 (0x0003)
R0x001E	seq_preview_1_fd	0000 0000 dddd dddd	2 (0x0002)
R0x001F	seq_preview_1_awb	0000 0000 dddd dddd	3 (0x0003)
R0x0020	seq_preview_1_hg	0000 0000 dddd dddd	3 (0x0003)
R0x0021	seq_preview_1_flash	0000 0000 dddd dddd	0 (0x0000)
R0x0022	seq_preview_1_skipframe	0000 0000 dddd dddd	0 (0x0000)
R0x0023	seq_preview_2_ae	0000 0000 dddd dddd	4 (0x0004)
R0x0024	seq_preview_2_fd	0000 0000 dddd dddd	0 (0x0000)
R0x0025	seq_preview_2_awb	0000 0000 dddd dddd	1 (0x0001)
R0x0026	seq_preview_2_hg	0000 0000 dddd dddd	1 (0x0001)
R0x0027	seq_preview_2_flash	0000 0000 dddd dddd	0 (0x0000)
R0x0028	seq_preview_2_skipframe	0000 0000 dddd dddd	16 (0x0010)
R0x0029	seq_preview_3_ae	0000 0000 dddd dddd	0 (0x0000)
R0x002A	seq_preview_3_fd	0000 0000 dddd dddd	0 (0x0000)
R0x002B	seq_preview_3_awb	0000 0000 dddd dddd	0 (0x0000)
R0x002C	seq_preview_3_hg	0000 0000 dddd dddd	0 (0x0000)
R0x002D	seq_preview_3_flash	0000 0000 dddd dddd	0 (0x0000)
R0x002E	seq_preview_3_skipframe	0000 0000 dddd dddd	64 (0x0040)

Table 16: Driver ID = 2: Auto Exposure Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0002	ae_window_pos	0000 0000 dddd dddd	0 (0x0000)
R0x0003	ae_window_size	0000 0000 dddd dddd	255 (0x00FF)
R0x0007	ae_gate	0000 0000 dddd dddd	4 (0x0004)
R0x000B	ae_min_index	0000 0000 dddd dddd	0 (0x0000)
R0x000C	ae_max_index	0000 0000 dddd dddd	24 (0x0018)
R0x000D	ae_min_virt_gain	0000 0000 dddd dddd	16 (0x0010)
R0x000E	ae_max_virt_gain	0000 0000 dddd dddd	128 (0x0080)
R0x0017	ae_status	0000 0000 0000 0ddd	0 (0x0000)
R0x0018	ae_current_y	0000 0000 ??? ???? ?	75 (0x004B)
R0x0019	ae_r12	???? ???? ???? ?	633 (0x0279)
R0x001B	ae_index	0000 0000 ??? ???? ?	4 (0x0004)
R0x001C	ae_virt_gain	0000 0000 ??? ???? ?	16 (0x0010)



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Table 16: Driver ID = 2: Auto Exposure Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0022	ae_r9	dddd dddd dddd dddd	16 (0x0010)
R0x002D	ae_r9_step	dddd dddd dddd dddd	157 (0x009D)
R0x004A	ae_target_min	0000 0000 dddd dddd	50 (0x0032)
R0x004B	ae_target_max	0000 0000 dddd dddd	150 (0x0096)
R0x004C	ae_target_buffer_speed	0000 0000 dddd dddd	12 (0x000C)
R0x004F	ae_base_target	0000 0000 dddd dddd	54 (0x0036)

Table 17: Driver ID = 3: Auto White Balance Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0002	awb_window_pos	0000 0000 dddd dddd	0 (0x0000)
R0x0003	awb_window_size	0000 0000 dddd dddd	239 (0x00EF)
R0x0006	awb_ccm_l_0	dddd dddd dddd dddd	435 (0x01B3)
R0x0008	awb_ccm_l_1	dddd dddd dddd dddd	65306 (0xFF1A)
R0x000A	awb_ccm_l_2	dddd dddd dddd dddd	51 (0x0033)
R0x000C	awb_ccm_l_3	dddd dddd dddd dddd	65421 (0xFF8D)
R0x000E	awb_ccm_l_4	dddd dddd dddd dddd	512 (0x0200)
R0x0010	awb_ccm_l_5	dddd dddd dddd dddd	65395 (0xFF73)
R0x0012	awb_ccm_l_6	dddd dddd dddd dddd	65434 (0xFF9A)
R0x0014	awb_ccm_l_7	dddd dddd dddd dddd	65306 (0xFF1A)
R0x0016	awb_ccm_l_8	dddd dddd dddd dddd	589 (0x024D)
R0x0018	awb_ccm_l_9	dddd dddd dddd dddd	28 (0x001C)
R0x001A	awb_ccm_l_10	dddd dddd dddd dddd	58 (0x003A)
R0x001C	awb_ccm_rl_0	dddd dddd dddd dddd	13 (0x000D)
R0x001E	awb_ccm_rl_1	dddd dddd dddd dddd	51 (0x0033)
R0x0020	awb_ccm_rl_2	dddd dddd dddd dddd	65498 (0xFFDA)
R0x0022	awb_ccm_rl_3	dddd dddd dddd dddd	13 (0x000D)
R0x0024	awb_ccm_rl_4	dddd dddd dddd dddd	65485 (0xFFCD)
R0x0026	awb_ccm_rl_5	dddd dddd dddd dddd	38 (0x0026)
R0x0028	awb_ccm_rl_6	dddd dddd dddd dddd	77 (0x004D)
R0x002A	awb_ccm_rl_7	dddd dddd dddd dddd	26 (0x001A)
R0x002C	awb_ccm_rl_8	dddd dddd dddd dddd	65434 (0xFF9A)
R0x002E	awb_ccm_rl_9	dddd dddd dddd dddd	12 (0x000C)
R0x0030	awb_ccm_rl_10	dddd dddd dddd dddd	65516 (0xFFEC)
R0x0032	awb_ccm_0	dddd dddd dddd dddd	442 (0x01BA)
R0x0034	awb_ccm_1	dddd dddd dddd dddd	65371 (0xFF5B)
R0x0036	awb_ccm_2	dddd dddd dddd dddd	65521 (0xFFFF1)
R0x0038	awb_ccm_3	dddd dddd dddd dddd	65479 (0xFFC7)
R0x003A	awb_ccm_4	dddd dddd dddd dddd	441 (0x01B9)
R0x003C	awb_ccm_5	dddd dddd dddd dddd	65415 (0xFF87)
R0x003E	awb_ccm_6	dddd dddd dddd dddd	65529 (0xFFFF9)
R0x0040	awb_ccm_7	dddd dddd dddd dddd	65330 (0xFF32)



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Table 17: Driver ID = 3: Auto White Balance Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0042	awb_ccm_8	dddd dddd dddd dddd	476 (0x01DC)
R0x0044	awb_ccm_9	dddd dddd dddd dddd	60 (0x003C)
R0x0046	awb_ccm_10	dddd dddd dddd dddd	43 (0x002B)
R0x0048	awb_gain_buffer_speed	0000 0000 dddd dddd	8 (0x0008)
R0x0049	awb_jump_divisor	0000 0000 dddd dddd	2 (0x0002)
R0x004A	awb_gain_min_r	0000 0000 dddd dddd	89 (0x0059)
R0x004B	awb_gain_max_r	0000 0000 dddd dddd	166 (0x00A6)
R0x004C	awb_gain_min_b	0000 0000 dddd dddd	89 (0x0059)
R0x004D	awb_gain_max_b	0000 0000 dddd dddd	166 (0x00A6)
R0x004E	awb_gain_r	0000 0000 dddd dddd	128 (0x0080)
R0x004F	awb_gain_g	0000 0000 dddd dddd	128 (0x0080)
R0x0050	awb_gain_b	0000 0000 dddd dddd	128 (0x0080)
R0x0051	awb_ccm_position_min	0000 0000 dddd dddd	0 (0x0000)
R0x0052	awb_ccm_position_max	0000 0000 dddd dddd	127 (0x007F)
R0x0053	awb_ccm_position	0000 0000 dddd dddd	64 (0x0040)
R0x0054	awb_saturation	0000 0000 dddd dddd	128 (0x0080)
R0x0055	awb_mode	0000 0000 dddd dddd	0 (0x0000)
R0x0056	awb_gain_r_buf	dddd dddd dddd dddd	0 (0x0000)
R0x0058	awb_gain_b_buf	dddd dddd dddd dddd	0 (0x0000)
R0x005D	awb_steady_b_gain_out_min	0000 0000 dddd dddd	120 (0x0078)
R0x005E	awb_steady_b_gain_out_max	0000 0000 dddd dddd	134 (0x0086)
R0x005F	awb_steady_b_gain_in_min	0000 0000 dddd dddd	126 (0x007E)
R0x0060	awb_steady_b_gain_in_max	0000 0000 dddd dddd	130 (0x0082)
R0x0065	awb_x0	0000 0000 dddd dddd	16 (0x0010)
R0x0066	awb_kr_l	0000 0000 dddd dddd	170 (0x00AA)
R0x0067	awb_kg_l	0000 0000 dddd dddd	150 (0x0096)
R0x0068	awb_kb_l	0000 0000 dddd dddd	128 (0x0080)
R0x0069	awb_kr_r	0000 0000 dddd dddd	145 (0x0091)
R0x006A	awb_kg_r	0000 0000 dddd dddd	130 (0x0082)
R0x006B	awb_kb_r	0000 0000 dddd dddd	130 (0x0082)

Table 18: Driver ID = 4: Anti-Flicker Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0002	fd_window_posh	0000 0000 dddd dddd	29 (0x001D)
R0x0003	fd_window_height	0000 0000 dddd dddd	4 (0x0004)
R0x0004	fd_mode	0000 0000 dd?d ????	0 (0x0000)
R0x0008	fd_search_f1_50	0000 0000 dddd dddd	30 (0x001E)
R0x0009	fd_search_f2_50	0000 0000 dddd dddd	32 (0x0020)
R0x000A	fd_search_f1_60	0000 0000 dddd dddd	37 (0x0025)
R0x000B	fd_search_f2_60	0000 0000 dddd dddd	39 (0x0027)
R0x000D	fd_stat_min	0000 0000 dddd dddd	3 (0x0003)



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Table 18: Driver ID = 4: Anti-Flicker Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x000E	fd_stat_max	0000 0000 dddd dddd	5 (0x0005)
R0x0010	fd_min_amplitude	0000 0000 dddd dddd	5 (0x0005)
R0x0011	fd_r9_step_f60_a	dddd dddd dddd dddd	157 (0x009D)
R0x0013	fd_r9_step_f50_a	dddd dddd dddd dddd	188 (0x00BC)
R0x0015	fd_r9_step_f60_b	dddd dddd dddd dddd	0 (0x0000)
R0x0017	fd_r9_step_f50_b	dddd dddd dddd dddd	224 (0x00E0)

Table 19: Driver ID = 7: Mode Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0003	mode_output_width_a	dddd dddd dddd dddd	640 (0x0280)
R0x0005	mode_output_height_a	dddd dddd dddd dddd	512 (0x0200)
R0x0007	mode_output_width_b	dddd dddd dddd dddd	1280 (0x0500)
R0x0009	mode_output_height_b	dddd dddd dddd dddd	1024 (0x0400)
R0x000D	mode_sensor_row_start_a	dddd dddd dddd dddd	0 (0x0000)
R0x000F	mode_sensor_col_start_a	dddd dddd dddd dddd	0 (0x0000)
R0x0011	mode_sensor_row_end_a	dddd dddd dddd dddd	1213 (0x04BD)
R0x0013	mode_sensor_col_end_a	dddd dddd dddd dddd	1613 (0x064D)
R0x0015	mode_sensor_row_speed_a	dddd dddd dddd dddd	8466 (0x2112)
R0x0017	mode_sensor_read_mode_a	dddd dddd dddd dddd	1132 (0x046C)
R0x0019	mode_sensor_fine_correction_a	dddd dddd dddd dddd	123 (0x007B)
R0x001B	mode_sensor_fine_it_min_a	dddd dddd dddd dddd	1032 (0x0408)
R0x001D	mode_sensor_fine_it_max_margin_a	dddd dddd dddd dddd	171 (0x00AB)
R0x001F	mode_sensor_frame_length_a	dddd dddd dddd dddd	659 (0x0293)
R0x0021	mode_sensor_line_length_pck_a	dddd dddd dddd dddd	2000 (0x07D0)
R0x0023	mode_sensor_row_start_b	dddd dddd dddd dddd	4 (0x0004)
R0x0025	mode_sensor_col_start_b	dddd dddd dddd dddd	4 (0x0004)
R0x0027	mode_sensor_row_end_b	dddd dddd dddd dddd	1211 (0x04BB)
R0x0029	mode_sensor_col_end_b	dddd dddd dddd dddd	1611 (0x064B)
R0x002B	mode_sensor_row_speed_b	dddd dddd dddd dddd	8465 (0x2111)
R0x002D	mode_sensor_read_mode_b	dddd dddd dddd dddd	36 (0x0024)
R0x002F	mode_sensor_fine_correction_b	dddd dddd dddd dddd	164 (0x00A4)
R0x0031	mode_sensor_fine_it_min_b	dddd dddd dddd dddd	1032 (0x0408)
R0x0033	mode_sensor_fine_it_max_margin_b	dddd dddd dddd dddd	164 (0x00A4)
R0x0035	mode_sensor_frame_length_b	dddd dddd dddd dddd	1261 (0x04ED)
R0x0037	mode_sensor_line_length_pck_b	dddd dddd dddd dddd	3334 (0x0D06)
R0x0039	mode_crop_x0_a	dddd dddd dddd dddd	0 (0x0000)
R0x003B	mode_crop_x1_a	dddd dddd dddd dddd	639 (0x027F)
R0x003D	mode_crop_y0_a	dddd dddd dddd dddd	0 (0x0000)
R0x003F	mode_crop_y1_a	dddd dddd dddd dddd	511 (0x01FF)
R0x0047	mode_crop_x0_b	dddd dddd dddd dddd	0 (0x0000)
R0x0049	mode_crop_x1_b	dddd dddd dddd dddd	1279 (0x04FF)
R0x004B	mode_crop_y0_b	dddd dddd dddd dddd	0 (0x0000)



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Table 19: Driver ID = 7: Mode Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x004D	mode_crop_y1_b	dddd dddd dddd dddd	1023 (0x03FF)
R0x0055	mode_output_format_a	dddd dddd dddd dddd	0 (0x0000)
R0x0057	mode_output_format_b	dddd dddd dddd dddd	0 (0x0000)
R0x0059	mode_spec_effects_a	dddd dddd dddd dddd	25664 (0x6440)
R0x005B	mode_spec_effects_b	dddd dddd dddd dddd	25664 (0x6440)
R0x005D	mode_y_rgb_offset_a	0000 0000 dddd dddd	0 (0x0000)
R0x005E	mode_y_rgb_offset_b	0000 0000 dddd dddd	0 (0x0000)
R0x0063	mode_common_mode_settings_fx_sepia_settings	dddd dddd dddd dddd	0 (0x0000)
R0x0066	mode_common_mode_settings_test_mode	0000 0000 dddd dddd	0 (0x0000)

Table 20: Driver ID = 11: Histogram Variables

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x0004	hg_max_dark_level	0000 0000 dddd dddd	64 (0x0040)
R0x0006	hg_black_clip_percent	0000 0000 dddd dddd	3 (0x0003)
R0x0008	hg_dark_level	0000 0000 dddd dddd	16 (0x0010)
R0x001B	hg_brightness_metric	dddd dddd dddd dddd	0 (0x0000)
R0x001F	hg_llmode	0000 0000 dddd dddd	132 (0x0084)
R0x0028	hg_ll_brightness_start	dddd dddd dddd dddd	5500 (0x157C)
R0x002A	hg_ll_brightness_stop	dddd dddd dddd dddd	7000 (0x1B58)
R0x0037	hg_gamma_morph_ctrl	0000 0000 dddd dddd	3 (0x0003)
R0x0038	hg_gamma_start_morph	dddd dddd dddd dddd	5500 (0x157C)
R0x003A	hg_gamma_stop_morph	dddd dddd dddd dddd	7000 (0x1B58)
R0x003C	hg_gamma_table_a_0	0000 0000 dddd dddd	0 (0x0000)
R0x003D	hg_gamma_table_a_1	0000 0000 dddd dddd	27 (0x001B)
R0x003E	hg_gamma_table_a_2	0000 0000 dddd dddd	46 (0x002E)
R0x003F	hg_gamma_table_a_3	0000 0000 dddd dddd	76 (0x004C)
R0x0040	hg_gamma_table_a_4	0000 0000 dddd dddd	120 (0x0078)
R0x0041	hg_gamma_table_a_5	0000 0000 dddd dddd	152 (0x0098)
R0x0042	hg_gamma_table_a_6	0000 0000 dddd dddd	176 (0x00B0)
R0x0043	hg_gamma_table_a_7	0000 0000 dddd dddd	232 (0x00E8)
R0x0044	hg_gamma_table_a_8	0000 0000 dddd dddd	207 (0x00CF)
R0x0045	hg_gamma_table_a_9	0000 0000 dddd dddd	217 (0x00D9)
R0x0046	hg_gamma_table_a_10	0000 0000 dddd dddd	225 (0x00E1)
R0x0047	hg_gamma_table_a_11	0000 0000 dddd dddd	232 (0x00E8)
R0x0048	hg_gamma_table_a_12	0000 0000 dddd dddd	238 (0x00EE)
R0x0049	hg_gamma_table_a_13	0000 0000 dddd dddd	242 (0x00F2)
R0x004A	hg_gamma_table_a_14	0000 0000 dddd dddd	246 (0x00F6)
R0x004B	hg_gamma_table_a_15	0000 0000 dddd dddd	249 (0x00F9)
R0x004C	hg_gamma_table_a_16	0000 0000 dddd dddd	251 (0x00FB)
R0x004D	hg_gamma_table_a_17	0000 0000 dddd dddd	253 (0x00FD)
R0x004E	hg_gamma_table_a_18	0000 0000 dddd dddd	255 (0x00FF)



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Table 20: Driver ID = 11: Histogram Variables (continued)

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x004F	hg_gamma_table_b_0	0000 0000 dddd dddd	0 (0x0000)
R0x0050	hg_gamma_table_b_1	0000 0000 dddd dddd	15 (0x000F)
R0x0051	hg_gamma_table_b_2	0000 0000 dddd dddd	26 (0x001A)
R0x0052	hg_gamma_table_b_3	0000 0000 dddd dddd	46 (0x002E)
R0x0053	hg_gamma_table_b_4	0000 0000 dddd dddd	80 (0x0050)
R0x0054	hg_gamma_table_b_5	0000 0000 dddd dddd	106 (0x006A)
R0x0055	hg_gamma_table_b_6	0000 0000 dddd dddd	128 (0x0080)
R0x0056	hg_gamma_table_b_7	0000 0000 dddd dddd	145 (0x0091)
R0x0057	hg_gamma_table_b_8	0000 0000 dddd dddd	161 (0x00A1)
R0x0058	hg_gamma_table_b_9	0000 0000 dddd dddd	175 (0x00AF)
R0x0059	hg_gamma_table_b_10	0000 0000 dddd dddd	187 (0x00BB)
R0x005A	hg_gamma_table_b_11	0000 0000 dddd dddd	198 (0x00C6)
R0x005B	hg_gamma_table_b_12	0000 0000 dddd dddd	208 (0x00D0)
R0x005C	hg_gamma_table_b_13	0000 0000 dddd dddd	217 (0x00D9)
R0x005D	hg_gamma_table_b_14	0000 0000 dddd dddd	226 (0x00E2)
R0x005E	hg_gamma_table_b_15	0000 0000 dddd dddd	234 (0x00EA)
R0x005F	hg_gamma_table_b_16	0000 0000 dddd dddd	241 (0x00F1)
R0x0060	hg_gamma_table_b_17	0000 0000 dddd dddd	249 (0x00F9)
R0x0061	hg_gamma_table_b_18	0000 0000 dddd dddd	255 (0x00FF)

Table 21: GPIO Registers

1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic

Register #Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x1070	gpio_data	000d dddd 0000 0000	0 (0x0000)
R0x1074	gpio_output_set	000d dddd 0000 0000	0 (0x0000)
R0x1076	gpio_output_clear	000d dddd 0000 0000	0 (0x0000)
R0x1078	gpio_dir	000d dddd 0000 0000	7936 (0x1F00)



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Register Description

Table 22: Sensor Core Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name	Frame Sync'd	Bad Frame
R0x3012	15:0	0x0010	coarse_integration_time (R/W)	Y	N
			Integration time specified in multiples of line_length_pck_.		
R0x3014	15:0	0x00F9	fine_integration_time (R/W)	Y	N
			Integration time specified as a number of pixel clocks.		
R0x301E	15:0	0x002A	data_pedestal (RO)	N	Y
			Constant offset that is added to the ADC output for all visible pixels in order to set the black level to a value greater than 0. Read-only.		
R0x3028	15:0	0x000D	analog_gain_code_global (R/W)	Y	N
			Writing a gain code to this register is equivalent to writing that code to each of the 4 color-specific gain code registers. Reading from this register returns the value most recently written to the analog_gain_code_greenR register.		
R0x302A	15:0	0x000D	analog_gain_code_greenr (R/W)	Y	N
			The gain code written to this register sets the gain for green pixels on red/green rows of the pixel array.		
R0x302C	15:0	0x000D	analog_gain_code_red (R/W)	Y	N
			The gain code written to this register sets the gain for red pixels.		
R0x302E	15:0	0x000D	analog_gain_code_blue (R/W)	Y	N
			The gain code written to this register sets the gain for blue pixels.		
R0x3030	15:0	0x000D	analog_gain_code_greenb (R/W)	Y	N
			The gain code written to this register sets the gain for green pixels on blue/green rows of the pixel array.		
R0x3032	15:0	0x0100	digital_gain_greenr (R/W)	Y	N
			Digital gain applied to green pixels on red/green rows of the pixel array. The value is in unsigned 8.8 fixed-point format.		
R0x3034	15:0	0x0100	digital_gain_red (R/W)	Y	N
			Digital gain applied to red pixels of the pixel array. The value is in unsigned 8.8 fixed-point format.		
R0x3036	15:0	0x0100	digital_gain_blue (R/W)	Y	N
			Digital gain applied to blue pixels of the pixel array. The value is in unsigned 8.8 fixed-point format.		
R0x3038	15:0	0x0100	digital_gain_greenb (R/W)	Y	N
			Digital gain applied to green pixels on blue/green rows of the pixel array. The value is in unsigned 8.8 fixed-point format.		



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Table 22: Sensor Core Registers (continued)
R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name	Frame Sync'd	Bad Frame
R0x304A	15:0	0x0000	otpm_control (R/W)		
	15:7	X	Reserved		
	6	RO	otpm_control_auto_rd_success Indicates whether the automatic read sequence was successful.	N	N
	5	RO	otpm_control_auto_rd_end Indicates whether the automatic read sequence is finished.	N	N
	4	0x0000	otpm_control_auto_rd_start Trigger automatic OTP memory read sequence. The 16-bit value stored in the OTP memory will be read and available in otpm_read_value.	N	N
	3	X	Reserved		
	2	RO	otpm_control_auto_wr_success Indicates whether the automatic write sequence was successful.	N	N
	1	RO	otpm_control_auto_wr_end Indicates whether the automatic write sequence is finished.	N	N
	0	0x0000	otpm_control_auto_wr_start Trigger automatic OTP memory write sequence. The high voltage must be available on the high voltage pad before this sequence is triggered. The 16-bit value in otpm_program_value will be written to the OTP memory.	N	N
	OTP memory control				
R0x304C	15:0	0x0000	otpm_program_value (R/W)	N	N
	16-bit value that will be programmed to the OTP memory when triggered by otpm_control[0].				
R0x304E	15:0	0x0000	otpm_read_value (RO)	N	N
	16-bit value read from the OTP memory. The value will automatically be read at startup. The value will only be valid if otpm_control[6] indicates that the memory was read successfully.				



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Table 23: SOC1 Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3210	15:0	0x01B0	color_pipeline_control (R/W)
	15:9	X	Reserved
	8	0x0001	bypass_scaler Enable Scaler. 0: Scaler is bypassed. 1: Scaler is enabled.
	7	0x0001	gamma_en Enable Gamma correction 0: Gamma correction is disabled. 1: Gamma correction is enabled.
	6	X	Reserved
	5	0x0001	en_ccm Enable color correction 0: Color correction is disabled. 1: Color correction is enabled.
	4	0x0001	en_ap Enable 2D aperture correction. 0: 2D aperture correction is disabled. 1: 2D aperture correction is enabled.
	3	0x0000	shading_correction_enable Enable pixel shading correction. All coefficients and other configuration settings (including other fields in this register) must be set up before enabling shading correction. 0: Pixel shading correction is disabled. 1: Pixel shading correction is enabled.
	2:0	X	Reserved
	Color pipeline control		
R0x321C	15:0	0x0003	output_control_status (R/W)
	15:8	RO	Reserved
	7	0x0000	fifo_input_control This bit determines the source for the output FIFO; so it is only used when output select (R0x321C[3:0] = 0x0). 0: SOC 1: Sensor Bypass
	6:4	X	Reserved
	3:0	0x0003	output_select 0x0000: FIFO Output Selected 0x0001: Sensor Bypass 0x0002: Reserved 0x0003: FIFO Bypass 0x0004: Walking 1's Test Pattern 0x0005–0x0007: Reserved
	Output control and status		